Chapter 4: Linear dependence and linear independence

In this part of the lecture notes we will introduce the notions of *linear dependence* and *linear independence*. Vectors $v_1, v_2, \cdots, v_n$ are linearly independent if and only if the zero vector can be written in only one way as a linear combination of the v_j (namely as the trivial linear combination with 0-coefficients).

We will show that vectors v_j are linearly dependent (that is, are *not* linearly independent) if and only if one of the v_i is a linear combination of the other vectors v_j, that is, if and only if we can remove one of the v_i from the collection without changing the span.

Thus we can consider lists of vectors with as few elements as possible which span a given vector space V. Such "minimal" lists are automatically linearly independent. These considerations will lead into the theory of bases and dimension to be developed later.

Section 4.1: Linear dependence

In short, forming linear combinations is a way of producing new vectors from old.

Suppose you have some vectors $v_1, v_2, \cdots, v_n$ of a vector space. A *linear combination* of these is any vector that can be obtained from the v_j by "multiplication by a scalar" and "addition of vectors".

That is, v is a linear combination of $v_1, v_2, \cdots, v_n$ if there are scalars (=real numbers) $\lambda_1, \lambda_2, \cdots, \lambda_n$ such that

$$v = \lambda_1 v_1 + \lambda_2 v_2 + \ldots + \lambda_n v_n \ .$$

1 Linear dependence: definition

Definition. V a vector space over $\mathbb{R}$

$v_1, v_2, \cdots, v_n \in V$ (possibly with repetitions)

The vectors $v_1, v_2, \cdots, v_n \in V$ are *linearly dependent* if there are scalars $\lambda_1, \lambda_2, \cdots, \lambda_n \in \mathbb{R}$, not all $\mathbf{0}$, such that

$$\lambda_1 v_1 + \lambda_2 v_2 + \ldots + \lambda_n v_n = \underline{o} \ .$$

Clarification: The phrase "not all 0" means that *at least one* of the scalars λ_j is non-zero. (Note also that "not all 0" and "all not 0" mean very different things!)

Example: if $v_1 = \underline{o}$ we can *choose* $\lambda_1 = 1$ and $\lambda_2 = 0$ and $\lambda_3 = 0$ and ... and $\lambda_n = 0$. Then

$$\lambda_1 v_1 + \lambda_2 v_2 + \ldots + \lambda_n v_n = 1\underline{o} + 0v_2 + \ldots + 0v_n = \underline{o}$$

so that $v_1, v_2, \cdots, v_n$ are linearly dependent.

2 Linear dependence: what it means

The equality $\lambda_1 v_1 + \lambda_2 v_2 + \ldots + \lambda_n v_n = \underline{o}$ means that we write the zero vector $\underline{o}$ as a linear combination of the vectors $v_1, v_2, \cdots, v_n$. Of course we can always do this: simply choose $\lambda_1 = 0$ and $\lambda_2 = 0$ and ... and $\lambda_n = 0$. This is boring and tells us nothing about the vectors v_j.

In contrast, being "linearly dependent" means that we can write

$$\lambda_1 v_1 + \lambda_2 v_2 + \ldots + \lambda_n v_n = \underline{o}$$

in an "interesting" way, *with at least one of the coefficients* λ_j *non-zero*. This means we can write $\underline{o}$ as a *non-trivial* linear combination of the vectors $v_1, v_2, \cdots, v_n$.

The vectors $v_1, v_2, \cdots, v_n$ are **linearly dependent** if and only if we can write the zero-vector as a **non-trivial** linear combination of $v_1, v_2, \cdots, v_n$.

3 Linear dependence: Examples

The vectors $\begin{pmatrix} 2 \\ 8 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix} \in \mathbb{R}^3$ are linearly dependent since

$$1 \cdot \begin{pmatrix} 2 \\ 8 \\ 4 \end{pmatrix} \;+\; 0 \cdot \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} \;+\; (-2) \cdot \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix} \;=\; \underline{o} \;.$$

The vectors $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix}, \begin{pmatrix} 4 \\ 4 \\ 7 \end{pmatrix} \in \mathbb{R}^3$ are linearly dependent. Indeed,

$$2 \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix} - \begin{pmatrix} 4 \\ 4 \\ 7 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad .$$

How to find suitable scalars? Try with unknown coefficients:

$$\lambda_1 \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \lambda_2 \cdot \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix} + \lambda_3 \cdot \begin{pmatrix} 4 \\ 4 \\ 7 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

This gives a homogeneous system of three linear equations in the variables λ_1, λ_2 and λ_3, and we can find out if a non-trivial solution exists (Gauss elimination), and even get explicit solutions (for example, $\lambda_1 = 2$, $\lambda_2 = 1$ and $\lambda_3 = -1$).

4 Linear dependence: Examples (continued)

In $\mathcal{P}_3$, the vector space of polynomials of degree ≤ 3, the vectors

$$\begin{array}{rcrcrcrcrl} p_1 & = & x^3 & + & 2x^2 & & & + & 1 & , \\ p_2 & = & & & x^2 & - & x & - & 3 & , \\ p_3 & = & 2x^3 & + & 7x^2 & - & 3x & - & 7 & \end{array}$$

are linearly dependent.

The condition $\lambda_1 p_1 + \lambda_2 p_2 + \lambda_3 p_3 = \underline{o}$ leads to the homogeneous system of linear equations

$$\left.\begin{array}{lrcrcrcl} \text{(coefficient of } x^3\text{)} & \lambda_1 & & & + & 2\lambda_3 & = & 0 \\ \text{(coefficient of } x^2\text{)} & 2\lambda_1 & + & \lambda_2 & + & 7\lambda_3 & = & 0 \\ \text{(coefficient of } x^1\text{)} & & - & \lambda_2 & - & 3\lambda_3 & = & 0 \\ \text{(coefficient of } x^0\text{)} & \lambda_1 & - & 3\lambda_2 & - & 7\lambda_3 & = & 0 \end{array}\right\}$$

which has a non-trivial solution, *e.g.*, $\lambda_1 = 4$, $\lambda_2 = 6$, $\lambda_3 = -2$. That is, $4p_1 + 6p_2 - 2p_3 = \underline{o}$.

5 Linear dependence: two observations

Recall:

$v_1, v_2, \cdots, v_n$ *linearly dependent* $\Leftrightarrow$ have scalars λ_j, **not all zero**, with $\lambda_1 v_1 + \lambda_2 v_2 + \ldots + \lambda_n v_n = \underline{o}$.

If $v_k = \underline{o}$ for some index k, then $v_1, v_2, \cdots, v_n$ are linearly dependent.

For we can choose $\lambda_j = 0$ for $j \neq k$, and $\lambda_k = 1$:

$$0v_1 + \ldots + 0v_{k-1} + \mathbf{1}v_k + 0v_{k+1} + \ldots + 0v_n = \underline{o}$$

If a vector occurs twice in $v_1, v_2, \cdots, v_n$, then these vectors are linearly dependent.

For if $v_k = v_\ell$ for some $k < \ell$, then we can choose $\lambda_k = 1$, $\lambda_\ell = -1$, and $\lambda_j = 0$ for $j \neq k, \ell$:

$$0v_1 + \ldots + 0v_{k-1} + \mathbf{1v_k} + 0v_{k+1} + \ldots + (\mathbf{-1})\mathbf{v}_\ell + 0v_{\ell+1} + \ldots + 0v_n = \underline{o}$$

Section 4.2: Linear independence

6 Linear independence: definition

Definition. V a vector space over $\mathbb{R}$

$v_1, v_2, \cdots, v_n \in V$ (possibly with repetitions)

The vectors $v_1, v_2, \cdots, v_n \in V$ are *linearly independent* if the following is true:

Given any scalars $\mu_1, \mu_2, \cdots, \mu_n \in \mathbb{R}$ such that

$$\mu_1 v_1 + \mu_2 v_2 + \ldots + \mu_n v_n = \underline{o} \ , \tag{*}$$

then **we must have** $\mu_1 = \mu_2 = \ldots = \mu_n = 0$.

Of course all the μ_j *may* be zero, in which case (*) is true. The point is a different one: not assuming anything about the scalars μ_j initially, **if** (*) is true **then** all the μ_j **must** be 0.

7 Linear independence: what it means

The vectors $v_1, v_2, \cdots, v_n$ are linearly independent if the **only** way of writing the zero vector as a linear combination of the v_j is the trivial way (all coefficients 0).

So given vectors $v_1, v_2, \cdots, v_n$,

- either you **can** write $\underline{o}$ as a non-trivial linear combination of the v_j, in which case $v_1, v_2, \cdots, v_n$ are **linear dependent**,
- or you **cannot** write $\underline{o}$ as a non-trivial linear combination of the v_j, in which case $v_1, v_2, \cdots, v_n$ are **linear independent**.

The vectors $v_1, v_2, \cdots, v_n \in V$ are linearly independent if and only if they are *not* linearly dependent.

8 Linear independence: Example

In the $\mathbb{R}$-vector space $\mathcal{P}_2$, the vectors

$$p_1 = x^2 + x + 1 \ , \quad p_2 = x^2 + x + 2 \ , \quad p_3 = x^2 + 2x + 2$$

are linearly independent.

Why? Try to find μ_j with

$$\mu_1 p_1 + \mu_2 p_2 + \mu_3 p_3 = \underline{o}\ ;$$

writing this out leads to a system of linear equations:

$$\left.\begin{array}{lrcrcrcl}
\text{coefficient of } x^2 : & \mu_1 & + & \mu_2 & + & \mu_3 & = & 0 \\
\text{coefficient of } x : & \mu_1 & + & \mu_2 & + & 2\mu_3 & = & 0 \\
\text{constant coefficient :} & \mu_1 & + & 2\mu_2 & + & 2\mu_3 & = & 0
\end{array}\right\}$$

The **only** solution is $\mu_1 = \mu_2 = \mu_3 = 0$. Hence the vectors are linearly independent.

9 Linear independence: unit vectors in $\mathbb{R}^n$

Recall the unit vectors $e_j \in \mathbb{R}^n$: the jth component of e_j is 1, all other entries are 0.

The vectors $e_1, e_2, \cdots, e_n$ are linearly independent.

Proof. Try to find scalars μ_j with $\mu_1 e_1 + \mu_2 e_2 + \ldots + \mu_n e_n = \underline{o}$. Writing this out leads to a rather simple system of linear equations:

$$\left.\begin{array}{ccccrl}
\mu_1 & & & & = & 0 \\
& \mu_2 & & & = & 0 \\
& & \ddots & & \vdots & \\
& & & \mu_n & = & 0
\end{array}\right\}$$

The **only** solution is $\mu_1 = \mu_2 = \ldots = \mu_n = 0$. Hence the vectors are linearly independent. ∎

Section 4.3: Checking linear (in)dependence

10 Checking for linear (in)dependence amounts to solving a homogeneous system of linear equations

V an $\mathbb{R}$-vector space, $v_1, v_2, \cdots, v_n \in V$ given vectors

To check whether the vectors $v_1, \cdots, v_n$ are linearly dependent, try to find scalars $\lambda_1, \cdots, \lambda_n \in \mathbb{R}$ such that

$$\lambda_1 v_1 + \lambda_2 v_2 + \ldots + \lambda_n v_n = \underline{o}\ . \tag{$\dagger$}$$

Treat the λ_j as *variables*. In concrete cases ($V = \mathbb{R}^m$, or $V = \mathcal{P}_m$, or $V = \mathcal{M}_{m,k}$) the vector equation ($\dagger$) is equivalent to a homogeneous system of linear equations in n variables.

- If this system has the trivial solution $\lambda_1 = \lambda_2 = \ldots = \lambda_n = 0$ **only**, then the vectors $v_1, \cdots, v_n$ are *linearly independent*.
- If the system **has other solutions as well** (with at least one of the variables $\neq 0$), the vectors are *linearly dependent*.

11 Checking for linear (in)dependence: explicit example

Recall that $\mathcal{M}_{2,2}$ is the vector space of 2-by-2 matrices.

Let s be a real number.

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} \qquad B = \begin{pmatrix} 2 & s \\ 4 & 6 \end{pmatrix} \qquad C = \begin{pmatrix} 3 & 7 \\ 8 & 9 \end{pmatrix}$$

Question: For which value of s are the vectors A, B, C of $\mathcal{M}_{2,2}$ linearly dependent?

Try to work it out yourself before looking at the next slide!

12 Checking for linear (in)dependence: explicit example (continued 1)

The question is: *for which value of s can we write*

$$\alpha A + \beta B + \gamma C = \underline{o}$$

with at least one of α, β and γ non-zero?

Explicitly, the equation means

$$\alpha \cdot \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} + \beta \cdot \begin{pmatrix} 2 & s \\ 4 & 6 \end{pmatrix} + \gamma \cdot \begin{pmatrix} 3 & 7 \\ 8 & 9 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

And this translates into a system of linear equations:

$$\left.\begin{array}{rcrcrcl} \alpha & + & 2\beta & + & 3\gamma & = & 0 \\ \alpha & + & s\beta & + & 7\gamma & = & 0 \\ 2\alpha & + & 4\beta & + & 8\gamma & = & 0 \\ 3\alpha & + & 6\beta & + & 9\gamma & = & 0 \end{array}\right\}$$

13 Checking for linear (in)dependence: explicit example (continued 2)

A single step of Gauss elimination leads to the following equivalent system:

$$\left.\begin{array}{lrcrcrcl} & \alpha & + & 2\beta & + & 3\gamma & = & 0 \\ (II)-(I) & & & (s-2)\beta & + & 4\gamma & = & 0 \\ (III)-2(I) & & & & & 2\gamma & = & 0 \\ (IV)-3(I) & & & & & 0 & = & 0 \end{array}\right\}$$

If $s - 2 \neq 0$ then this system is in row echelon form. In this case, there is precisely one solution (no parameter), and since the system is homogeneous this must be the trivial solution $\alpha = \beta = \gamma = 0$.

If $s \neq 2$ the vectors A, B, C are linearly independent.

14 Checking for linear (in)dependence: explicit example (continued 3)

If however $s = 2$ then we need another Gauss elimination step to bring the system into row echelon form (note that $(s-2)\beta = 0$ in this case):

$$\begin{array}{rrrrrrrr} & \alpha & + & 2\beta & + & 3\gamma & = & 0 \\ & & & & & 4\gamma & = & 0 \\ (III) - \frac{1}{2}(II) & & & & & 0 & = & 0 \\ & & & & & 0 & = & 0 \end{array} \left.\vphantom{\begin{array}{c}1\\1\\1\\1\end{array}}\right\}$$

This system has one parameter (namely β). Hence there is a non-trivial solution!

If $s = 2$ the vectors A, B, C are linearly dependent.

15 Linear (in)dependence and linear combinations

Let V be an $\mathbb{R}$-vector space. The vectors $v_1,\ v_2,\ \cdots,\ v_n$ in V are linearly dependent if and only if one of them is a linear combination of the other vectors.

Proof. If $v_1, v_2, \ldots, v_n$ are linearly dependent, have scalars λ_j with

$$\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n = \underline{o}\ ,$$

and with $\lambda_i \neq 0$ for some index i. Bring $\lambda_i v_i$ to the other side and divide by $-\lambda_i$ to obtain

$$v_i = -\frac{\lambda_1}{\lambda_i} v_1 - \ldots - \frac{\lambda_{i-1}}{\lambda_i} v_{i-1} - \frac{\lambda_{i+1}}{\lambda_i} v_{i+1} - \ldots - \frac{\lambda_n}{\lambda_i} v_n\ .$$

This shows that v_i is a linear combination of the other vectors.

16 Linear dependence and linear combinations (continued)

Conversely, suppose v_j is a linear combination of the other vectors. Then we have scalars $\lambda_1, \cdots, \lambda_{j-1}, \lambda_{j+1}, \cdots, \lambda_n$ such that

$$v_j = \lambda_1 v_1 + \ldots + \lambda_{j-1} v_{j-1} + \lambda_{j+1} v_{j+1} + \ldots + \lambda_n v_n \ .$$

Bringing v_j to the other side,

$$\underline{o} = \lambda_1 v_1 + \cdots + \lambda_{j-1} v_{j-1} \mathbf{-1v_j} + \lambda_{j+1} v_{j+1} + v \cdots + \lambda_n v_n \ ,$$

and since not all coefficients in this linear combination are zero (the coefficient of v_j is -1; we don't know about the other λ_i!), we have shown that the vectors $v_1, \ldots, v_n$ are linearly dependent. ■

17 Linear dependence and linear combinations: Example 1 for $V = \mathbb{R}^2$

$$u = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \ , \quad w = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \ , \quad z = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

You can check that span $\{u, z\} = \mathbb{R}^2$. Thus we have $w \in$ span $\{u, z\}$. Hence u, w, z are linearly dependent by slide 15 of this chapter.

We *also* have $u \in$ span $\{w, z\}$ and $z \in$ span $\{u, w\}$, each showing (again) that u, w, z are linearly dependent.

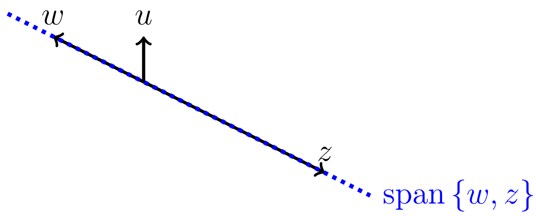
18 Linear dependence and linear combinations: Example 2 for $V = \mathbb{R}^2$

$$u = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \ , \quad w = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \ , \quad z = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$$

As $\operatorname{span}\{u, z\} = \mathbb{R}^2$ we have $w \in \operatorname{span}\{u, z\}$. Hence u, w, z are linearly dependent by slide 15 of this chapter. — We *also* have $z \in \operatorname{span}\{u, w\}$, however $u \notin \operatorname{span}\{w, z\}$ since

$$\operatorname{span}\{w, z\} = \{\left(\begin{smallmatrix} 2t \\ -t \end{smallmatrix}\right) \mid t \in \mathbb{R}\}$$

is a line (geometrically speaking) not containing u:



Section 4.4: Generating vector spaces by very few vectors

19 Generators of a vector space (reminder)

Recall: V is *finitely generated* if there are vectors $v_1, v_2, \cdots, v_n \in V$ with $\operatorname{span}\{v_1, v_2, \cdots, v_n\} = V$. In this case we call $\{v_1, v_2, \cdots, v_n\}$ a *generating set* for V.

Suppose that $S = \{v_1, v_2, \cdots, v_n\}$ is a generating set for the vector space V. Then so is the set $S' = \{v_1, v_2, \cdots, v_n, w_1, \cdots, w_\ell\}$, for *any* choice of vectors w_j.

So we can always make generating sets larger if we want to.

Proof. We have $V = \operatorname{span}(S) \subseteq \operatorname{span}(S') \subseteq V$ since S is generating, and by monotonicity of span (slide 19 of Chapter 3). Hence $\operatorname{span}(S') = V$. ∎

20 Minimal lists of vectors generating V

Let V be an $\mathbb{R}$-vector space. If the vectors $v_1, v_2, \cdots, v_n \in V$ generate V,

$$\text{span}\{v_1, v_2, \cdots, v_n\} = V \ ,$$

we *may* be able to delete one of the vectors so that the remaining $n-1$ vectors still generate V. We *may* then be able to delete another of the vectors so that the remaining $n-2$ vectors still generate V. We can go on like this: drop one of the vectors from the list again and again, always ensuring the remaining vectors still generate V, until you cannot delete another vector. Then the list is a "minimal" generating list:

Definition. We say that $v_1, v_2, \cdots, v_n \in V$ form a *minimal list of vectors generating* V if these n vectors generate V, but after deleting one of the v_j the remaining $n-1$ vectors do *not* generate V.

21 Minimality implies linear independence

Suppose that $v_1, v_2, \cdots, v_n \in V$ is a minimal list of vectors generating V. Then these vectors are linearly independent.

Proof. We prove the equivalent "contrapositive" statement:

> **If $v_1, v_2, \cdots, v_n$ are linearly dependent, we can delete one of the v_j so that the remaining vectors still generate V.**

$v_1, v_2, \cdots, v_n$ linearly dependent implies that one of the vectors is in the span of the others, see slide 15 of this chapter. So there is an index i such that $v_i \in \text{span}\{v_1, v_2, \cdots, v_{i-1}, v_{i+1}, \cdots, v_n\}$. But then

$$V = \text{span}\{v_1, v_2, \cdots, v_n\} = \text{span}\{v_1, v_2, \cdots, v_{i-1}, v_{i+1}, \cdots, v_n\}$$

by "omitting redundant generators" (slide 36 of Chapter 3). That is, we can delete the ith vector and still generate V, so the list wasn't minimal to begin with. ∎

22 Linear independence implies minimality

Conversely, if you omit a vector from a list of linearly independent vectors, you make the span smaller:

Let $v_1, v_2, \cdots, v_n$ be linearly independent vectors in an $\mathbb{R}$-vector space V. Then for any index k with $1 \le k \le n$ we have

$$\operatorname{span}\{v_1, v_2, \cdots, v_{k-1}, v_{k+1}, \cdots, v_n\} \neq \operatorname{span}\{v_1, v_2, \cdots, v_n\}.$$

(Of course the inclusion "$\subset$" holds, by monotonicity of span.)

Proof. We prove the contrapositive statement. For given k with $1 \le k \le n$, suppose the two spans are equal. Then

$$v_k \in \operatorname{span}\{v_1, v_2, \cdots, v_n\} = \operatorname{span}\{v_1, v_2, \cdots, v_{k-1}, v_{k+1}, \cdots, v_n\}$$

so v_k is in the span of the other vectors. Hence $v_1, v_2, \cdots, v_n$ are linearly dependent, see slide 15 of this chapter. ■

23 Summary

- Vectors $v_1, v_2, \cdots, v_n$ are *linearly independent* if and only if the following is true:

  The zero-vector $\underline{o}$ can be written as a linear combination of the v_j **in only one way** (*viz.*, the trivial linear combination with 0-coefficients).

- Vectors $v_1, v_2, \cdots, v_n$ are *linearly dependent* if and only if one of the v_i is in the span of the other v_j.
- If $v_1, v_2, \cdots, v_n$ is a shortest-possible list of vectors spanning V, then $v_1, v_2, \cdots, v_n$ are linearly independent.
- Conversely, if $v_1, v_2, \cdots, v_n$ are linearly independent then removing any of the vectors will make their span smaller:

$$\operatorname{span}\{v_1, v_2, \cdots, v_{n-1}\} \neq \operatorname{span}\{v_1, v_2, \cdots, v_{n-1}, v_n\}$$